

3D Box Detection and Measurement

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This report summarizes the implementation and evaluation of a simple box detection and measurement pipeline based on RANSAC plane fitting and mask-based refinement. The approach assumes a single box placed on a flat surface, with both top and bottom planes approximately parallel.

Implementation Summary

1. Point cloud data are loaded from Kinect `.mat` files and preprocessed.
2. The floor plane is detected using RANSAC, and inliers are visualized.
3. Non-floor points are isolated to detect the top plane of the box.
4. Binary masks for the floor and top are refined morphologically.
5. Box corners are extracted and 3D dimensions (width, length, height) are computed assuming top and floor planes are parallel.

Weaknesses and Limitations

- Single-box assumption: The algorithm expects only one object above the floor. Multiple boxes or cluttered environments cause detection errors.
- Parallel plane assumption: Height computation assumes top and bottom planes are parallel. Tilted boxes introduce bias.
- Threshold sensitivity: The RANSAC distance threshold (`RANSAC_DIST_THRESH`) critically affects inlier selection. Low thresholds miss data; high thresholds accept outliers.
- Noise and missing depth data: Kinect noise, invalid pixels, and low amplitude regions can disrupt plane fitting.
- Processing time: RANSAC with many iterations can be slow for dense point clouds.

Suggestions for Improvement

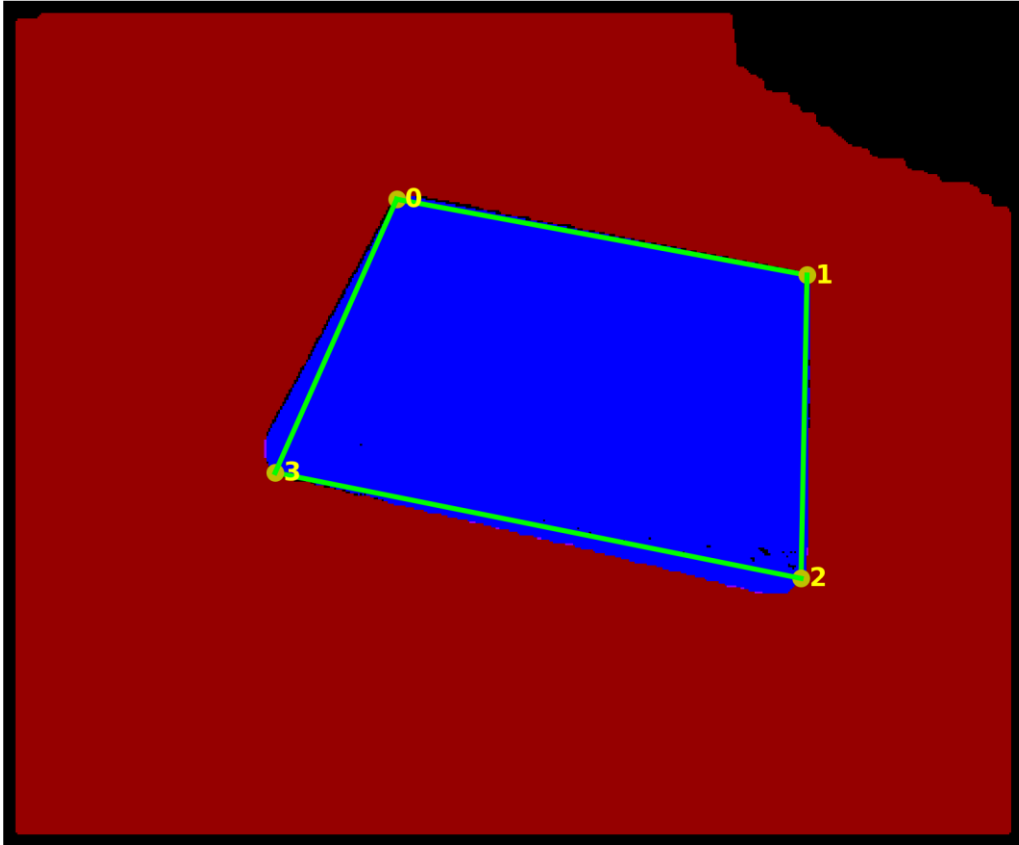
- Use region-growing or MLESAC instead of pure RANSAC for more stable plane estimation.
- Implement multi-object segmentation to handle multiple boxes.
- Employ parallelization (e.g., GPU or multiprocessing) for faster RANSAC iterations.
- Add adaptive thresholds that depend on local noise statistics.
- Introduce planar consistency checks to verify that top and bottom planes are parallel within tolerance.
- Include confidence metrics to quantify measurement uncertainty.

Box Measurement Method

The top mask contour is approximated to four polygon corners, which are then mapped to 3D space. Width and length are computed from corner distances, while height is derived from the distance between the floor and top planes. This method assumes clean segmentation and uniform box geometry.

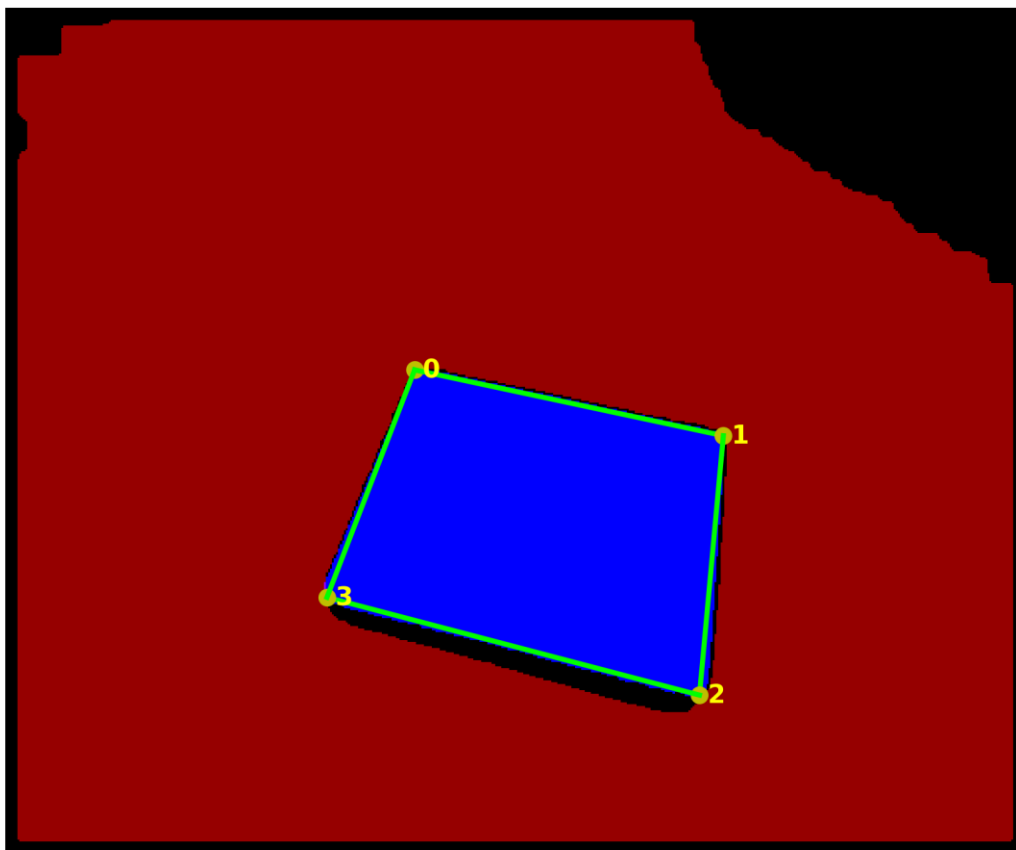
Results

$H=1.214$, $W=0.459$, $L=0.342$



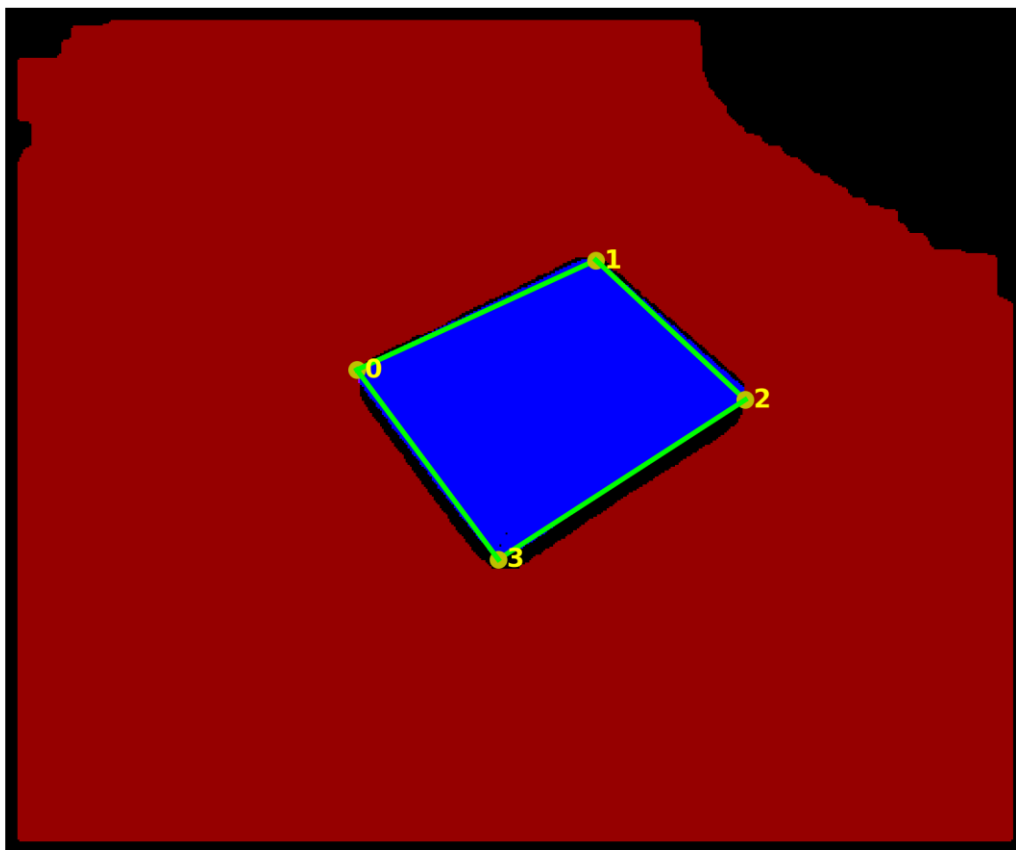
Example 1 – $H=1.214$, $W=0.459$, $L=0.342$

H=0.174, W=0.451, L=0.352



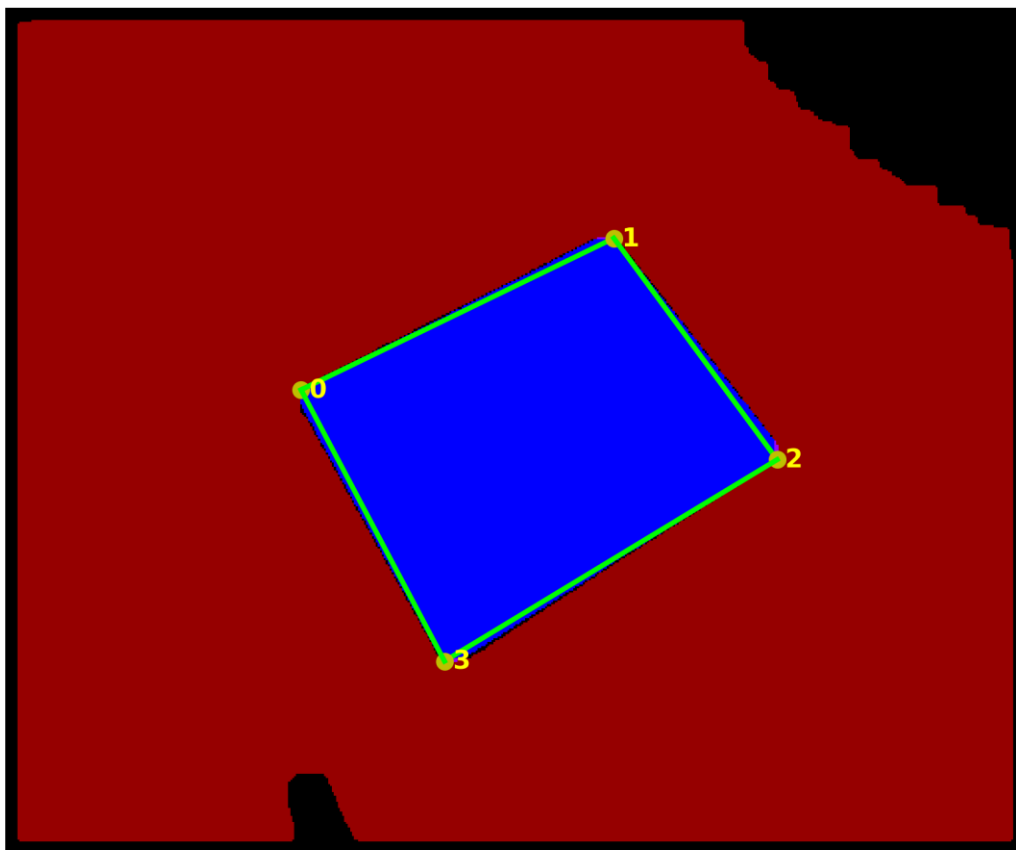
Example 2 – $H=0.174$, $W=0.451$, $L=0.352$

$H=1.888, W=0.435, L=0.337$



Example 3 – $H=1.888, W=0.435, L=0.337$

$H=0.158, W=0.448, L=0.352$



Example 4 – $H=0.158, W=0.448, L=0.352$